Lesson 13.2 - Correlation Coefficient

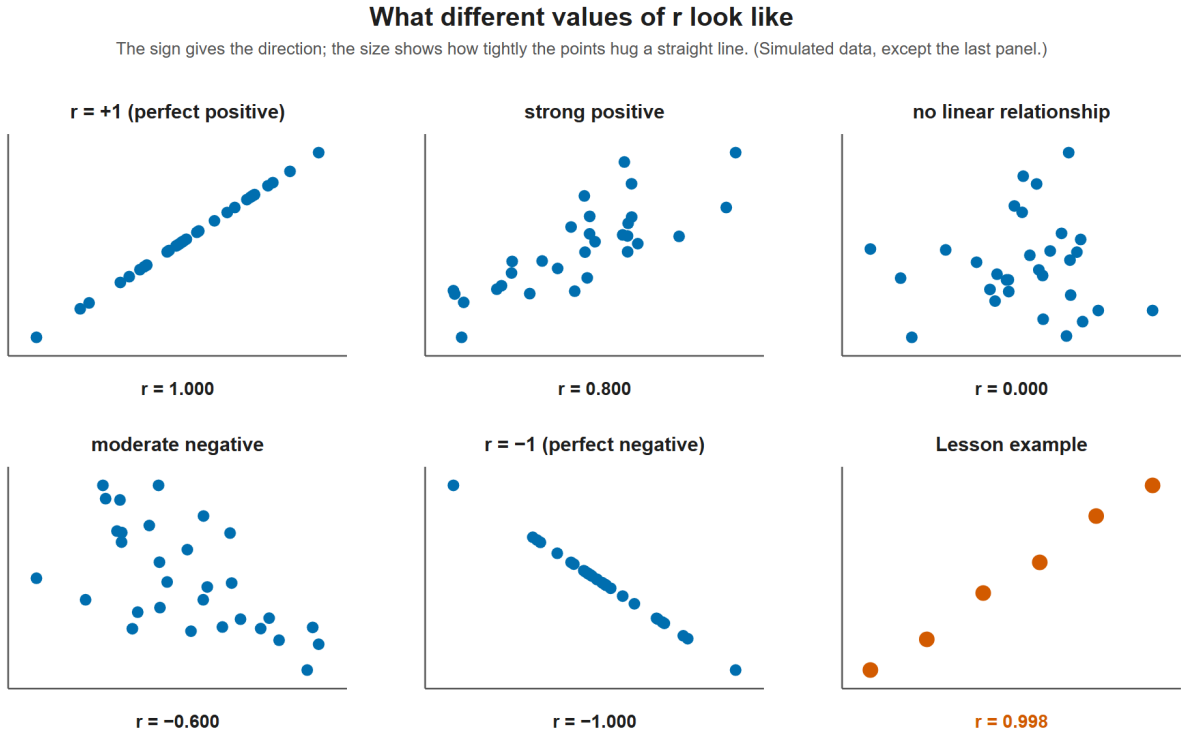
> **Video:** Correlation Coefficient by The Organic Chemistry Tutor **Chapter:** 13 - Regression & Correlation

What r tells you

The **correlation coefficient** r (sometimes written ρ) measures the **strength and direction of the linear relationship** between two variables. $-1 \leq r \leq 1$.

r	Picture	Relationship
$+1$	All points exactly on a line going **up**	Perfect positive (direct): y rises as x rises
-1	All points exactly on a line going **down**	Perfect negative (inverse): y falls as x rises
close to ± 1 (e.g. 0.8)	Points close to a line	Strong linear relationship
smaller (e.g. 0.6-0.7)	Points more scattered around the line	Weaker linear relationship
≈ 0	Random cloud of points	No linear relationship

The **sign** of r matches the sign of the slope. The **size** of r shows how tightly the points hug the line.



The closer r is to ±1, the more tightly the points line up. The sign follows the direction of the trend. The lesson's data (orange) is almost a perfect line, r = 0.998.

Formula

$$r = \frac{n\sum xy - \sum x \sum y}{\sqrt{\left[n\sum x^2 - (\sum x)^2\right]\left[n\sum y^2 - (\sum y)^2\right]}}$$

n = number of (x, y) pairs.

Example

x	y	xy	x^2	y^2
1	2	2	1	4
2	4	8	4	16
3	7	21	9	49
4	9	36	16	81
5	12	60	25	144
6	14	84	36	196
Sum	**48**	**211**	**91**	**490**

$\sum x = 21,\ \sum y = 48,\ \sum xy = 211,\ \sum x^2 = 91,\ \sum y^2 = 490,\ n = 6$

Numerator: $n\sum xy - \sum x \sum y = 6(211) - 21(48) = 1266 - 1008 = 258$

Denominator:

- $n\sum x^2 - \left(\sum x\right)^2 = 6(91) - 21^2 = 546 - 441 = 105$
- $n\sum y^2 - \left(\sum y\right)^2 = 6(490) - 48^2 = 2940 - 2304 = 636$
- $\sqrt{105 \times 636} = \sqrt{66{,}780} \approx 258.42$

$$r = \frac{n\sum xy - \sum x \sum y}{\sqrt{\left[n\sum x^2 - \left(\sum x\right)^2\right]\left[n\sum y^2 - \left(\sum y\right)^2\right]}} = \frac{258}{258.42} \approx 0.998$$

A **very strong positive** linear relationship: as x increases, y increases almost exactly along a straight line.

Key takeaways

- r ranges from −1 to +1. The sign gives the direction and the size gives the strength of the **linear** relationship.
- Build a table of x, y, xy, x^2, y^2, add each column, then use the formula.
- $r \approx 0$ means no *linear* relationship. There could still be a curved one.
- Correlation is not causation.